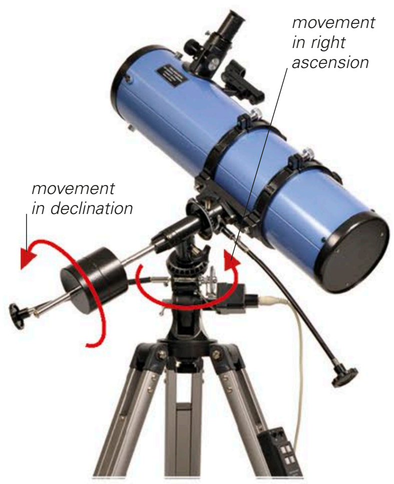
TELESCOPE MOUNTS

The way a telescope is mounted can greatly affect its performance. The two most common types of mount are the altazimuth and the equatorial. The altazimuth mount allows the instrument to pivot in altitude (up and down) and azimuth (parallel to the horizon). The equatorial mount aligns the telescope's movement with Earth's axis of rotation, so that it can follow the lines of right ascension and declination in the sky (see p.63). Altazimuth mountings are simple to set up, but because objects in the sky are constantly changing their altitude and azimuth, tracking objects requires continued adjustment of both. Equatorial mounts are heavier and take longer to set up but, once aligned to a celestial pole, the observer can follow objects across the sky by turning a single axis.

ALTAZIMUTH MOUNT
This type of mount is usually light and compact. However, both axes of the telescope must be moved at the same time to track a celestial object—and the higher the telescope's magnification, the faster the object will drift out of the field of view.



EQUATORIAL MOUNT
These mounts are more awkward to set up, but once that is done the observer can track objects just by turning the polar axis. Many equatorial mounts have electric or battery-controlled drive motors that allow for hands-free operation.

FORK MOUNT

DOBSONIAN MOUNT

ALTAZIMUTH VARIATIONS
There are two variants of the altazimuth mount. Fork mounts are often used for catadioptric telescopes. Dobsonians are good for large reflectors with wide fields of view and low magnifications.

APERTURE AND MAGNIFICATION

Two major factors affect an image in a telescope eyepiece—aperture and magnification. The aperture is the diameter of the telescope's primary mirror or objective lens and affects the amount of light it can collect—called its "light grasp." Doubling the aperture quadruples the light grasp. Magnification is dictated by the specification of the telescope's eyepiece. The power of the eyepiece is identified by its focal length—the distance at which it focuses parallel rays of light. The shorter the focal length, the greater the magnification. Objective lenses and primary mirrors also have a focal length, and dividing this measurement by that of the eyepiece gives the combined magnification. An eyepiece can be changed to alter the magnification to suit the observed object.

120mm APERTURE   66mm APERTURE

OBJECTIVE SIZES
The most important specification of a telescope is the diameter of its objective lens. This affects how much light can enter the tube.

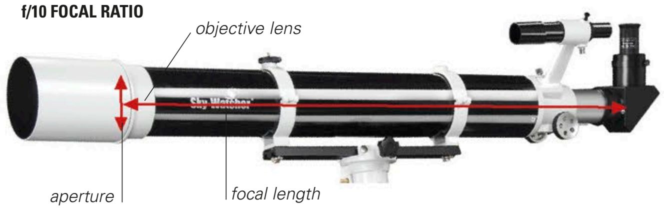
50mm APERTURE   100mm APERTURE

APERTURE
These are photographs of the open cluster M35. The image far left was taken through a telescope with a 2 in (50 mm) objective lens; the second image, left, was taken through a 4 in (100 mm) lens. The larger lens has a light grasp four times greater than the smaller one, so the fainter stars can be seen more clearly.

9mm EYEPIECE   25mm EYEPIECE

MAGNIFICATION
The shorter the focal length of an eyepiece, the higher its magnifying power but also the smaller its field of view. This can be seen clearly in these two photographs of the Moon. The image far left was taken through a 9 mm eyepiece; the second image was taken through a 25 mm eyepiece.

f/10 FOCAL RATIO



f/5 FOCAL RATIO

VARIATIONS IN FOCAL RATIO
Telescopes with a large focal ratio, such as f/10, above, produce larger images but have smaller fields of view than telescopes with lower focal ratios.

FOCAL LENGTH AND FOCAL RATIO

After the aperture, the next most important specification of a telescope is its focal length. This is the distance from its primary lens or mirror to the point where the rays of light meet—the focal point. A telescope with a long focal length produces a large but faint image at its focal point, whereas one with a shorter focal length gives a smaller but brighter image. It is easier to make mirrors with short focal lengths than it is lenses, so reflecting telescopes can have shorter tubes for a given aperture. Dividing the focal length of the primary mirror or lens (usually given in millimeters) by the telescope's aperture (also in millimeters) will give its focal ratio, called its "f" number. This ratio can influence the type of celestial object observed. Telescopes with a low focal ratio, around f/5, are best for imaging diffuse objects, such as nebulae or galaxies; those with a focal ratio above f/9 are useful for studying brighter objects, such as the Moon or the planets.